INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 1: Introduction and Research Problem

by

TAEHONG KWON

Student number: 28576810

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SUPERVISOR: Prof. Ernest Mnkandla

CO-SUPERVISOR: Prof. Donatien Koulla Moulla

CO-SUPERVISOR: Dr. David Sena Attipoe

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1 Introduction and Research Problem

Decentralized Finance (DeFi) has transformed how capital is allocated on public blockchains: lending, borrowing, and leveraged trading can now run without traditional intermediaries (Schär, 2021). Most credit-like transactions are still heavily collateralized, though, because protocols cannot draw on off-chain identity frameworks or existing credit data (Harvey et al., 2021). Under pseudonymity, the central issue is how to allocate capital efficiently among unknown parties.

This thesis asks what ERC-20 allowance edges add to PageRank as a measure of on-chain wallet reputation when one walk follows them together with transfers. The allowance edges come from the `Approval` events that `approve` and `permit` emit. The method of the thesis, a coupled PageRank (C-PR), walks the allowance edges and the transfer edges of the same addresses in one chain and mixes the two at each address with a fixed weight λ. Two single-layer walks serve as its references: EndorseRank, PageRank on the allowance edges alone with the edge weights of the transfer-based Adaptive Weighted PageRank (AWP) of T. Do et al. (2023), and AWP itself on the transfers. All of them are run on the same Arbitrum One contract accounts, beside the raw degrees that the walks smooth, and evaluated by the rank-correlation method used with AWP (T. Do et al., 2023), extended with paired contrasts, raw-degree baselines and temporal holdouts. In this thesis, an on-chain reputation metric is a value derived from on-chain activity and interactions without any off-chain identity information (H.-D. Do & Do, 2023).

1.1 Background: DeFi, collateralization, and pseudonymity

Public blockchains make programmable settlement possible without a central authority (Nakamoto, 2008). A smart contract can encode the terms of a deal (Szabo, 1997), and a DeFi platform that uses such contracts clears trades on the blockchain instead of routing them through a custodian

(Buterin, 2014). Protocols then stack these contracts into decentralized markets for lending, automated market making and perpetual futures (Werner et al., 2022). On Arbitrum One and other L2 networks, fees are low enough that ordinary users transact often, and every transfer, approval and protocol event can be inspected wallet by wallet.

Even with this visibility, DeFi credit markets remain inherently over-collateralized. Most protocols make a borrower or trader post more collateral than the position is worth, because the counterparty is a pseudonymous account whose owner has not been verified (Gudgeon, Perez, et al., 2020; Harvey et al., 2021). Pseudonymity is a basic property of public ledgers. Anyone can create addresses at almost no cost, and one entity can own more than one wallet (Douceur, 2002). Conventional credit ratings depend on personal histories that translate poorly to this setting (Packin & Lev-Aretz, 2024), so protocols control risk through collateral reserves, liquidation mechanisms, and protocol rules instead of personalized reputation.

Collateralization protects protocols, but it uses capital inefficiently. Traders and borrowers have to tie up funds they could otherwise invest, and the platforms offer none of the under-collateralized products that traditional finance extends to people with a credit record. Researchers have therefore turned to on-chain signals as a substitute for off-chain credit histories: transfer-graph centrality (H.-D. Do & Do, 2023), network risk propagation (D. Lin et al., 2025), prospect theory and blockchain-based credit scoring (Hassija et al., 2020), and machine learning pipelines over lending protocols (Kotb, 2023). Each of these lines of work assumes that public ledger data contains information usable for risk assessment without deanonymizing users.

Graph ranking algorithms appear well suited to this problem. PageRank and its variants rank nodes by recursive endorsement, so that a node's score depends on the endorsements it receives from other nodes (Page et al., 1999). Earlier web trust algorithms such as EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) apply similar ideas to peer-to-peer trust and to web spam. H.-D. Do and Do (2023) use both PageRank and HodgeRank on a graph of transfers to judge how trustworthy Ethereum users are, and call the result a form of social credit. AWP (T. Do et al., 2023) is another transfer-graph ranking algorithm. It weights each transfer by a bounded function of its

value and a logistic function of its age, and it restarts the walk at addresses that sent recently; its authors validate it by showing that it correlates with inbound transfer count and value, taken as proxies for economic importance. These studies establish transfer-graph reputation as a credible baseline, but they leave out explicit spending authorization, a separate on-chain action recorded as ERC-20 allowances (Lundfall, 2020; Vogelsteller & Buterin, 2015).

1.2 Motivation for on-chain reputation

Four main factors drive the need for on-chain reputation research in leveraged DeFi.

- Capital efficiency: protocols are seeking ordinal ranking signals that could support tiered services (more frequent limit increases, lower fees, or higher priority in monitoring, for example), and perhaps even differentiated collateral requirements in future controlled experiments, while settlement stays on chain, where the contract enforces collateralization ratios and liquidation conditions (Schär, 2021).
- Auditability: off-chain credit scores are opaque, but graph-based scores computed from public event data can be recomputed and audited, which would help to ease some of the concerns about black-box decentralized credit scoring (Packin & Lev-Aretz, 2024).
- Operational cost: a score that is refreshed often, before each leverage adjustment for instance, should be cheap to recompute.
- Standardization: other markets use screening ratios to tell counterparties apart, but pseudonymous DeFi has no standard ordinal language for doing so at the wallet level. Chapter 5 discusses where the scores studied here could serve that role and where they cannot.

Transfer-based AWP is public and auditable, but its graph grows with every payment (T. Do et al., 2023). Allowance graphs are a structurally different alternative. Whenever a user approves an ERC-20 token, the contract records how much of that token a spender may transfer (Lundfall, 2020; Vogelsteller & Buterin, 2015), and each new approval supersedes the older ones, so the latest approval of each owner–spender pair and token stands for the history of that pair. Approvals are

also rarer than transfers. On the same contracts the endorsement graph G_{approve} therefore has fewer than half as many edges as the transfer graph G_{transfer}, and the PageRank solve, whose cost grows linearly with $|E|$, takes correspondingly less time (Table 4.1). The same solver runs in both cases, so the speedup comes from the sparser graph and is not an algorithmic improvement; Claim C5 treats it as an operational property of the representation.

Efficiency aside, allowances also mean something different. A transfer might reflect routing, market making, or a passive receipt, whereas an allowance is an active grant of permission to spend funds. Treating allowances as endorsement edges carries PageRank's citation reading (Page et al., 1999) over to a trust relationship that protocols already depend on. Whether this produces different rankings, whether those rankings agree differently with authorization and transfer checks, and whether one walk over both kinds of edge ranks contracts better than a walk over either, is examined on a matched cohort of $n = 15{,}107$ contract accounts on Arbitrum One.[1]

1.3 Research problem

AWP infers trust from transfer history through weighted sums and logistic time-decay (T. Do et al., 2023). It (i) must aggregate the whole transfer history, (ii) models a high-value wallet as one that receives many inbound transfers, and (iii) assigns no role to spending authorization. EndorseRank, the allowance reference of this thesis, keeps AWP's edge weights and solver but builds the directed graph from the latest allowance of each owner–spender pair and token. The resulting graph is sparser, encodes a different signal and ranks a different side of each relationship. An allowance points from the owner to the spender, so EndorseRank scores the spenders that receive approvals, such as routers and vaults, and gives every owner that receives none the same value. Each of the two walks therefore follows one kind of act: AWP the payments that reach an address, EndorseRank the spending rights it is granted.

[1] Matched means that every method scores the same contracts: those approved by at least three distinct owners on WETH, USDC, USDT0, ARB or WBTC between 1 October 2023 and 30 June 2026, a list drawn from the approval logs alone (Section 3.2).

Because both edge types live on the same addresses, a single walk can follow both, mixing them node by node with a fixed weight λ. This walk, the Coupled PageRank (C-PR), is the method of the thesis; at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. The mixture is worth testing because the two edge types record different acts on the same contracts: an operator that already runs a transfer walk could add allowance edges to it instead of running a second score. The main research question is: what does a single PageRank walk over both ERC-20 allowance and transfer edges add to ranking contract accounts on Arbitrum One, compared with walks over either edge type and the raw degrees they smooth? Research questions RQ1–RQ5 (Section 1.6) break it down. The analysis plan fixed one decision rule, and its object is C-PR (Section 3.7).

Agreement with statistics of a score's own edges, or prediction of more of the same edges, does not yet show that a score measures reputation in a substantive sense. A comparison registered with the second holdout window, which adds no research question, therefore also uses an outcome built from neither edge type and observed after the freeze: the later liquidation avoidance of the GMX V2 accounts that are contracts, in small samples (Section 3.7.3); it is applied to the revised scores as a re-analysis.

The scores appear in Table 1.1. Table 1.2 lists hypotheses (H1–H3), research questions (RQ1–RQ5), and empirical claims (C1–C6) from Chapter 4.

The main terms are defined as follows:

- *DeFi (Decentralized Finance):* Financial services such as lending and borrowing, implemented as smart contracts on public blockchains and generally run without a central operator in charge of custody or settlement (Schär, 2021).
- *Smart contract:* Self-executing program code that runs on a blockchain and carries out an agreement without a trusted intermediary (Szabo, 1997).
- *Wallet / Address:* A blockchain account that holds funds and interacts with smart contracts, identified by its address (Buterin, 2014). It is either an externally owned account (EOA), a user's wallet controlled by a private key, or a contract account governed by its code

Table 1.1: The scores evaluated in this thesis (matched contract cohort, $n = 15{,}107$).

Score	Input graph	Role
C-PR	Both layers, US-dollar amounts with each allowance capped, mixed per node with weight λ	The method of this thesis, object of rules A and B. Its layers walked alone, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), are weighted by amounts, so they are not EndorseRank and AWP. Its ablation S-PR restarts a transfer walk from the allowance walk
EndorseRank	Latest ERC-20 allowances of five tokens in US dollars (G_{approve}), weighted as in AWP	Allowance delegation; the single-layer reference on allowances. Its variant with restarts at approving owners is reported beside it
AWP	Transfers of five tokens in US dollars (G_{transfer})	Transfer flow; the single-layer reference on transfers, the published transfer-graph method of T. Do et al. (2023) (Section 3.6.1)
Raw degrees	G_{approve} and G_{transfer}, counted per node	In-approve degree and transfer in-degree, the local counts that the PageRanks smooth: proxies in the same window and baselines out of window

(Antonopoulos & Wood, 2018). The wallets ranked here are contract accounts; EOAs enter only as owners, senders and recipients. No off-chain identity is assumed or recovered.

- *On-chain reputation score:* A number computed from on-chain data that expresses a wallet's relative position for financial activity under pseudonymity (H.-D. Do & Do, 2023). Here the scored addresses are contracts, and the only off-chain inputs are the five-token list and daily US-dollar prices. Scores are ordinal and meant for ranking; their absolute scale has no meaning for credit limits.
- *PageRank:* A graph ranking algorithm that spreads influence through a directed graph with a damping factor (Page et al., 1999). Adaptive Weighted PageRank (AWP) is PageRank on the transfer graph, with edges weighted by the value and age of each transfer and restarts at recently active senders (T. Do et al., 2023); here values are in US dollars, and the value weight is one half at the median amount. EndorseRank applies AWP's edge weights to the latest-allowance graph and restarts uniformly (Section 3.6).
- *Coupled PageRank (C-PR):* The method of this thesis: one random walk over both the al-

lowance and the transfer layer, which at each wallet mixes the two rows with weights λ and $1 - \lambda$ over the layers that wallet occupies (Section 3.6.3). Endorsement-seeded PageRank (S-PR) keeps the transfer walk but restarts it from the allowance walk, as TrustRank (Gyöngyi et al., 2004) restarts from a seed set.

- *ERC-20 allowance / Approve / Permit:* Contract methods that record how many of the owner's tokens a spender may spend on the owner's behalf (Lundfall, 2020; Vogelsteller & Buterin, 2015). `Approve` is an ordinary on-chain call; EIP-2612 `permit` makes the same state change through a signature. For each owner–spender pair and token, EndorseRank uses the latest non-zero allowance.
- *Domain alignment:* The degree to which a reputation ranking agrees with an observable proxy ranking, measured with Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938) and read as evidence on construct validity (Cronbach & Meehl, 1955); with proxies built from a score's own edges it is only an intended-construct check.

1.4 Temporal holdout

The contemporaneous allowance Kendall τ (Table 4.3) is an intended-construct check: PageRank on G_{approve} should agree with spender-side approval degree, because it smooths that degree globally. To test whether the score conveys information beyond the same-window graph, the scores are frozen at t_1 (31 March 2026, 23:59:59 UTC) and compared with two labels that only become available in April–June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). In this first window, W0, the holdout cohort ($n = 17{,}050$) consists of every contract spender that holds at least one positive latest allowance of one of the five tokens at t_1 in the extracted approval data. Each PageRank is also compared with the raw degree it smooths at t_1.

A second window, W1, was registered at 16:49 UTC on 9 October 2026, after the W0 results were known and before the study extracted any log dated after 30 June 2026. Its scores are frozen on

30 June 2026 and its labels counted from July to September 2026. It repeats the two spender labels on the $n = 18{,}152$ contract spenders of that freeze and adds one for the GMX V2 traders that are contracts ($n = 70$): the share of their closes that did not end in liquidation (Section 3.7.2).
The analysis plan of 9 October 2026 and the registration of W1 define the registered analysis, with amounts in raw base units of every token. After its results were known, the amounts were revised on 10 October 2026 to US dollars on five widely traded tokens (Section 3.5.1). The revised analysis is the main one; its verdicts under the rules of the registered analysis are a re-analysis, not registered tests, and the registered analysis reached the same verdicts (Appendix 8.6).

1.5 Research objectives

The research has five objectives, one for each of RQ1–RQ5. Each is stated with the measurement used to operationalize it and with the criterion that Chapter 5 applies when judging whether the objective was met.

O1. Establish whether EndorseRank and AWP, the two single-layer walks, order the contracts of the same cohort differently, and assess how closely each score agrees with each construct in the same window (RQ1). Measurement: Kendall τ between the two score vectors ($n = 15{,}107$) with a 95% bootstrap interval, and family-level Kendall τ with 95% intervals for every score and construct. Criteria: the interval of the agreement between the two walks contains neither 0 (unrelated orderings) nor 1 (identical orderings); a score agrees with a construct when its interval on that family excludes 0, and agreement with the construct of the score's own edges is read as an intended-construct check, because those proxies come from the same events as the graph.

O2. Measure what each score costs to compute on the same contracts, C-PR on the union of both graphs included, and how runtime responds to the number of sampled contracts (RQ2). Measurement: solve time (mean and standard deviation of five timed runs after one warm-up run), memory, iterations, $|V|$ and $|E|$, all under one protocol. Criterion: the time ratio is

Table 1.2: Summary of hypotheses, research questions, and empirical claims.

Item	Addresses	Reported in
H1, H2 / RQ1	Do EndorseRank and AWP, the walks over allowance edges alone and over transfer edges alone, order the same contracts differently, and does each agree most with the checks of its own edge type?	§4.3 and §4.4; Tables 4.3 and 4.4; Claim C4
H3 / RQ2	What does each walk cost to compute at equal n, and how much does walking both edge sets add?	Tables 4.1 and 4.2; Claim C5
RQ3	Do t_1 scores predict t_2 new approvals and new senders, and does any walk, the coupled one included, do better than the raw degree of the same edge type?	§4.6 and §4.7.4; Tables 4.9, 4.10, 4.11, 4.12, 4.15, 4.13 and 4.18; Claims C2 and C3
RQ4	Does C-PR rank contracts better than either of its layers walked alone, in window and out of window (rule A)?	§4.7.1 and §4.7.2; Tables 4.3, 4.14 and 4.10; Claim C1
RQ5	Does adding the allowance layer improve a transfer walk on new data (rule B)?	§4.7.3; Tables 4.15 and 4.16; Claim C1
C1	Under rules A and B, applied to the revised scores, C-PR beats neither layer walked alone and does not improve on its transfer layer within the margin (the gain on new approval pairs holds, non-inferiority fails on new senders and narrowly on the liquidation-free close rate); against AWP it fails clearly; in S-PR allowance edges help mainly when they carry the walk	Tables 4.14, 4.10 and 4.16
C2	Beside the rules, C-PR ranks both spender labels above EndorseRank in both windows and new approval pairs above AWP only through the spenders AWP scores zero; it ranks new senders below AWP and the transfer in-degree, and new approval pairs below EndorseRank with activity restarts (contrasts computed after the labels were known)	Table 4.18; §4.7.4
C3	Out of window, AWP ranks later new senders above the transfer in-degree in both windows, while EndorseRank ranks later new approval pairs below the in-approve degree in both (contrasts computed after the labels were known)	Tables 4.9, 4.10, 4.11 and 4.13
C4	Same-window agreement of each single-layer walk with the checks of its own edge type, partly built in, and different orderings of the cohort by the two walks	Tables 4.3 and 4.4; Figure 4.3
C5	Runtime and graph size of each score: speed follows graph size, not the algorithm; C-PR and S-PR are slower than AWP	Table 4.1
C6	On later liquidation avoidance of GMX V2 contract traders, no count-level lead of allowance over transfer in the registered window (positive in W0); PageRank versions not resolved; approval count lower on the prof	Tables 4.19 and 4.20

reported only together with the ratio of $|E|$, so that any difference can be traced to the size of the graph.

O3. Test whether scores frozen at t_1 predict new approvals and new transfer senders at t_2 among the cohort's spenders, and whether any PageRank, the coupled one included, does better than the raw degree of the same edge type (RQ3). Measurement: Kendall τ with 95% intervals between every score computed at the freeze, the two raw degrees included, and both labels, in W0 (freeze 31 March 2026, labels April–June 2026, $n = 17{,}050$ contract spenders) and W1 (freeze 30 June 2026, labels July–September 2026, $n = 18{,}152$), with paired contrasts of each single-layer PageRank against its own degree and of C-PR against both degrees. Criterion: a score predicts a label when its interval excludes 0, and a PageRank improves on a degree only when the interval of the paired difference lies above 0. Neither the degree contrasts nor this criterion was fixed before the W0 labels were known.

O4. Test whether one walk across both edge types beats each of its layers walked alone (RQ4). Measurement: C-PR at $\lambda = 0.5$ (with 0.25 and 0.75 as sensitivity values) and its ablation S-PR, compared through paired intervals on $\Delta\tau$ with the two layers walked alone, on the holdout and same-window labels. Criterion: rule A, fixed in the analysis plan of 9 October 2026 before any log of the study was extracted, under which C-PR does no worse than the allowance walk on new approvals and better than the transfer walk on new senders in W0, and in the same window lies within the interval of the leading layer on the transfer and allowance families and above both layers on Sybil stability. The rule is applied to the revised scores as a re-analysis.

O5. Test whether adding the allowance layer to a walk over the transfers improves it on new data (RQ5). Measurement: paired contrasts of C-PR with its transfer layer walked alone in W1, on new approval pairs, new senders and the traders' liquidation-free close rate. Criterion: rule B, registered on 9 October 2026 after the W0 results were known and before the study extracted the W1 logs, under which C-PR beats the transfer walk on new approval pairs in W1 and is not worse by more than 0.01 on new senders or on the traders' liquidation-free

close rate. The rule is applied to the revised scores as a re-analysis. A registered sensitivity analysis repeats it with AWP in place of the transfer walk; it does not change the rule. The contrasts of C-PR with EndorseRank, AWP and the other scores are reported beside both rules and decide neither.

1.6 Research questions

Three hypotheses go with the first two research questions. H1: on the same set of contracts, EndorseRank and AWP produce different orderings. H2: each score agrees with the checks built from the edges it walks on. H3: at the same value of n, a score's computation time grows with the size of its graph, so EndorseRank, on the sparsest graph, is the fastest to compute. H1–H3 hold largely by construction and act as checks of the instruments. The falsifiable tests are RQ3–RQ5, the last two answered by rules fixed before the study extracted the relevant data and re-applied to the revised scores (Section 3.7).

RQ1. On the same set of Arbitrum contracts, do EndorseRank and AWP, the walks over allowance edges alone and over transfer edges alone, produce different orderings, and how closely does each score agree with current allowance checks and with current transfer checks (Spearman's ρ and Kendall's τ)?

RQ2. What does each score cost to compute on the same set of contracts, as measured by time, memory, and iterations, and how much does walking both edge sets add?

RQ3. Given scores computed at time t_1, do the scores predict t_2 new owner–spender approvals to inbound spenders and new transfer senders, and does any PageRank, the coupled one included, predict them better than the raw degree of the same edge type at t_1?

RQ4. Does a coupled PageRank running on both the allowance and the transfer layer rank contracts better than a walk over either layer alone, in sample as well as out of sample?

RQ5. Does adding the allowance layer to a walk over the transfers, such as AWP, improve it out of sample?

RQ3 uses later labels of the same two constructs because no default or repayment label exists for the contracts ranked here (Section 2.8). RQ1 and RQ2 establish the two single-layer walks and what they cost; RQ4 and RQ5 move from comparing the two edge types to combining them. The rule for RQ5, rule B, came after the W0 labels had been seen, so it is judged only on W1 (Sections 3.7 and 3.7.2).

1.7 Summary of the literature review

Chapter 2 reviews four areas. In graph ranking, PageRank (Page et al., 1999) ranks nodes by recursively propagated endorsement at a per-iteration cost linear in the number of edges (Langville & Meyer, 2006), and EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) adapt it to peer-to-peer trust and web spam with normalization and seed sets. In on-chain reputation, H.-D. Do and Do (2023) apply PageRank and HodgeRank to Ethereum transactions, and AWP (T. Do et al., 2023) weights transaction edges by value and age and checks its ranking against inbound transfer count and value, statistics drawn from the same data as its graph. D. Lin et al. (2025) propagate account risk through the transaction network, and surveys of blockchain trust and reputation systems (Bellini et al., 2020) and of reputation systems built on smart contracts (Almasoud et al., 2020) describe designs that rest on transfers or explicit ratings. Deployed DeFi tooling such as Human Passport instead aggregates on-chain and off-chain credentials to decide whether an address belongs to a distinct person (Human Passport, n.d.). None of this work takes the ERC-20 allowance (Vogelsteller & Buterin, 2015) as a ranking edge; address clustering uses approvals only to link addresses of one entity (Victor, 2020). In DeFi risk and measurement, systematizations classify risk by protocol layer (Werner et al., 2022) and by the attacks seen in practice (Zhou et al., 2023), and wash trading (Cong et al., 2023; Victor & Weintraud, 2021), extractable value (Qin et al., 2022), scam tokens (Xia et al., 2021) and airdrop farming (Luo et al., 2025) show that much transfer activity is artificial. In measurement theory, construct validity (Cronbach & Meehl, 1955; Messick, 1995) separates what a score measures from external criteria, and the bootstrap (DiCiccio & Efron,

1996; Efron, 1979) gives distribution-free intervals for rank statistics. The gap (Section 2.21) is a walk that couples allowance and transfer edges, evaluated on one cohort beside the walks over each edge type and the raw degrees, with confidence intervals, paired differences, a fixed timing protocol and temporal holdouts.

1.8 Summary of the research methodology

Chapter 3 applies a quantitative, comparative design to public ledger data. ERC-20 `Approval` and `Transfer` logs on Arbitrum One from 1 October 2023 to 30 September 2026 are extracted from Google BigQuery in two scans, the second after W1 was registered. The analysis keeps the events of five tokens around the matched cohort ($n = 15{,}107$), valued in US dollars as revised after the registered results were known (Section 3.5.1). One power-iteration PageRank implementation scores the coupled walk over both layers, the latest-allowance graph and the transfer graph. Each score is compared with Spearman's ρ and Kendall's τ against same-window transfer, allowance and Sybil-stability proxies and, out of window, against new approval pairs and new transfer senders on the spender contracts of W0 and W1, with the raw degrees at each freeze as baselines. Except in the value-slope and sample-definition checks, every coefficient has a 95% interval from 400 paired contract resamples (seed 42), and method comparisons are paired. A comparison registered with W1 sets allowance-side against transfer-side scores on the later liquidation avoidance of GMX V2 contract traders in both windows (2,000 paired resamples; Bonferroni 97.5% intervals for its two decision contrasts). The code, configuration and machine-readable summaries behind every table are in a public repository (Section 3.13), and the study uses only public data (Section 3.16).

1.9 Contributions

The thesis makes five contributions to the study of on-chain reputation and DeFi risk. EndorseRank and AWP both apply weighted PageRank (Page et al., 1999) to a single edge type, each with its own restart distribution; the coupled walk C-PR changes the transition rule itself. What is new is

therefore a transition rule over two on-chain relations, its test under rules fixed in advance, an edge type, an evaluation design, and an analytic account of what such a walk computes on this sample.

- Coupled walk over authorization and payment (C-PR): one random walk over a US-dollar allowance layer, with each allowance capped, and a US-dollar transfer layer, which at each address mixes the two transition rows with a fixed weight λ over the layers that address occupies; S-PR is its seeded ablation. No reputation method reviewed couples the two relations (Section 2.21).
- Its test under fixed rules: rule A, fixed with λ in the analysis plan, asked C-PR in its primary part to do no worse than its allowance layer on new approval pairs and better than its transfer layer on new senders, and rule B, registered with W1, asked it to improve on its transfer layer; only these contrasts change the coupling and nothing else. On the revised scores, in a re-analysis, both fail as in the registered analysis (Appendix 8.6), though the gain over the transfer layer on new approval pairs recurs in W1 (Claim C1). Beside the rules, C-PR ranks both spender labels above EndorseRank in both windows, but it ranks later senders below AWP and later approval pairs below EndorseRank with activity restarts, and its lead over AWP on later approvals comes from contracts that AWP cannot order (Claim C2). The ablation shows that the allowance signal survives when allowance edges carry the walk and is mostly lost when they only decide where it restarts.
- Latest-allowance edge construction (EndorseRank), the allowance reference: an endorsement graph in which an edge $u \to v$ stands for the latest in-window ERC-20 allowance, set by `approve` or `permit`, that the spender v holds from the owner u, valued in US dollars and weighted as AWP weights a transfer. The edge records a delegation of spending rights, not a history of payments. Approvals have been used to cluster the addresses of one entity (Victor, 2020), but no reputation method reviewed ranks on them (Section 2.21).
- Same-cohort evaluation with out-of-window tests and degree baselines: every score is computed on one set of contracts and checked against identical transfer, allowance and Sybil-

stability proxies under the protocol used with AWP (T. Do et al., 2023), with 95% bootstrap intervals for the coefficients and paired intervals for the contrasts (Efron, 1979); scores frozen at t_1 are tested against the new approvals and new transfer senders that arrive after t_1, beside the raw degrees that PageRank smooths, which separates in-sample agreement from prediction and the propagated score from its local count (Claims C2 and C3). In contrasts computed after the labels were known, AWP beat its degree in both windows, EndorseRank did so only with AWP's activity restarts, a hypothesis for a registered test, and C-PR ranked neither label above the degree of the same edge type. A comparison registered with W1 adds an outcome built from neither edge type: on later liquidation avoidance the approval count showed no lead over the transfer count in the registered window, though it led in W0, and the like-for-like PageRank pair is not resolved, in small trader samples (Claim C6).

- Closed form on a depth-one graph: with uniform restarts and uniformly redistributed dangling mass, a spender whose in-neighbors have no in-edges scores $c\,(1 + d\,\delta(v))$, where c is the common score of nodes without in-edges, d the damping factor and $\delta(v)$ the sum of the transition probabilities into v; in EndorseRank this is the owner-normalized in-strength, which divides each owner's endorsement over all the spenders it approves. The proposition needs only the uniform restarts, so it describes C-PR's walk as well. On EndorseRank's graph it holds to within 1% for every depth-one spender but for only a minority of the deeper ones, so multi-hop propagation is present (Section 5.3.2). Besides the time and token weights, that normalization and this propagation separate EndorseRank from the raw approval count, but over all spenders they did not make it rank later approvals above that count, and deeper spenders were not tested apart.

1.10 Scope and limitations

The study covers Arbitrum One from 2023-10-01 to 2026-06-30, and W1 adds July to September 2026. Its main sample is the matched cohort of $n = 15{,}107$ contracts, and its holdout samples

the contract spenders of each freeze ($n = 17{,}050$ in W0, $n = 18{,}152$ in W1). The scored wallets are contract accounts, because user wallets serve too many different purposes for their activity to be put in one order (Section 3.2); account type is read from the code at a pinned block, not at the time of each event. Only events of the five tokens with a cohort contract at one end enter the graphs, so each graph is the ego network of the cohort, and a multi-hop path is seen only where each of its edges touches a cohort contract (Section 5.3.2). Few GMX V2 contract accounts in the freeze-date graph closed at least three positions in these windows ($n = 88$ in W0, $n = 70$ in W1), so every trader result has wide intervals (Section 4.8). Runtime is also measured on subsamples of the cohort. No claim is made that the results generalize beyond Arbitrum One or the months studied. Amounts are in US dollars, a choice made after the registered results were known (Section 3.5.1); EndorseRank and AWP pass them through a bounded function that gives the median amount half weight, while the layers of C-PR use the amounts themselves, with each allowance capped (Chapter 3). Adversarial manipulation of allowances or transfers is a real threat, because addresses are cheap to create (Douceur, 2002): Section 5.6 shows that a small farm of Sybil addresses can lift a contract's score on the allowance graph above most of the cohort cheaply, while reaching the top is expensive, that the bound behind this holds for C-PR as well, and that the Sybil-adjusted proxies describe some manipulation patterns without protecting the score against them.

1.11 Research roadmap

Chapter 2 reviews graph-based on-chain reputation, ERC-20 allowance semantics and the reputation tooling deployed in DeFi, and develops the theoretical framework. Chapter 3 defines the cohorts, the tokens, prices and the revision of the amounts, the graphs and scores, the construct checks, the temporal holdouts, the timing protocol, and the standards for reproducibility and ethics. Chapter 4 reports the benchmark (Claim C5), the same-window alignment of every score and the divergence of the two single-layer walks (C4), robustness checks, the temporal holdouts of the single-layer walks (C3), the coupled walk under its rules and beside every other score (C1 and C2)

and the comparison on liquidation avoidance (C6); Appendix 8.5 collects the direct comparisons between EndorseRank and AWP, and Appendix 8.6 reports the registered analysis. Chapter 5 interprets the findings by research question and objective, works out their implications, models the cost of Sybil attacks on allowance graphs, and weighs threats to validity and limitations. Chapter 6 answers RQ1–RQ5, states the contribution to knowledge and sets out future work.